Cursors in MySQL – Exercise 18

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1. CREATE THE TABLE

   CREATE TABLE employees (id INTEGER, Name VARCHAR(100), email VARCHAR(100));

   Output:

```
Query OK, 0 rows affected (0.012 sec)
```

2. INSERT RECORDS

   INSERT INTO employees(id, Name, email) VALUES (1, 'Harry Potter', 'pharry@warnerbros.com');
   INSERT INTO employees(id, Name, email) VALUES (2, 'Clark Kent', 'kclark@dccomics.com');
   INSERT INTO employees(id, Name, email) VALUES (3, 'Tony Stark', 'stony@marvel.com');

   Output:

```
Query OK, 1 row affected (0.030 sec)

Query OK, 1 row affected (0.022 sec)

Query OK, 1 row affected (0.011 sec)
```

3. VIEW THE EMPLOYEES TABLE

   SELECT * FROM employees;

   Output:

```
+----+--------------+----------------------+
| id | Name         | email                |
+----+--------------+----------------------+
|  1 | Harry Potter | pharry@warnerbros.com |
|  2 | Clark Kent   | kclark@dccomics.com   |
|  3 | Tony Stark   | stony@marvel.com      |
+----+--------------+----------------------+
3 rows in set (0.002 sec)
```

4. CREATE THE PROCEDURE WITH A CURSOR

```
DELIMITER $$
CREATE PROCEDURE build_email_list (INOUT email_list VARCHAR(4000))
BEGIN
    DECLARE v_finished INTEGER DEFAULT 0;
    DECLARE v_email VARCHAR(100) DEFAULT '';

    -- declare cursor for employee email
    DECLARE email_cursor CURSOR FOR
        SELECT email FROM employees;

    -- declare NOT FOUND handler
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET v_finished = 1;

    OPEN email_cursor;

    get_email: LOOP
        FETCH email_cursor INTO v_email;
        IF v_finished = 1 THEN
            LEAVE get_email;
        END IF;
        -- build email list
        SET email_list = CONCAT(v_email, ';', email_list);
    END LOOP get_email;

    CLOSE email_cursor;
END$$
DELIMITER ;
```

Output:

```
Query OK, 0 rows affected (0.034 sec)
```

5. CALL THE PROCEDURE AND GET THE EMAIL LIST

```
SET @email_list = '';
CALL build_email_list(@email_list);
SELECT @email_list;
```

Output:

```
Query OK, 0 rows affected (0.000 sec)

Query OK, 0 rows affected (0.004 sec)

+-------------------------------------------------------------+
| @email_list                                                 |
+-------------------------------------------------------------+
| stony@marvel.com;kclark@dccomics.com;pharry@warnerbros.com; |
+-------------------------------------------------------------+
1 row in set (0.000 sec)
```